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Embedded packaging structure and manufacturing method thereof

Abstract

A multilayer embedded packaging structure according to an embodiment includes a first dielectric layer and a second dielectric layer on the first dielectric layer. The first dielectric layer includes a first wiring layer. The second dielectric layer includes a first copper pillar layer and a device placement port frame penetrating through the second dielectric layer in a height direction, and a second wiring layer on the first copper pillar layer. A second copper pillar layer is on the second wiring layer. The first wiring layer and the second wiring layer are conductively connected via the first copper pillar layer. A first device is mounted to the bottom of the device placement port frame, a second device is mounted to the second dielectric layer, and a third device is mounted to an end of the second copper pillar layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to Chinese patent application no. 202110294274.6 filed on Mar. 19, 2021 in the Chinese intellectual property office, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

1. Technical Field

(2) The present invention relates to a packaging structure for an electronic device, and in particular to an embedded packaging structure and a manufacturing method thereof.

2. Description of the Related Art

(3) With the rapid development of electronic technology, electronic products spring up, related electronic products go further and further on the road of intellectualization and integration, and the miniaturization and integration of related electronic products become the mainstream of the market. The miniaturization and intellectualization mean that each electronic component constituting the core part of the electronic products demands higher requirements, and how to manufacture lower cost, faster speed, more reliable performance and smaller dimension are the goals of integrated circuit packaging.

(4) Semiconductor front-end wafer manufacture changes from the original tens of nanometers to more than ten nanometers to even fewer than ten nanometers with the technology continuing to expand to make various breakthroughs. The integrated circuit packages are ushering into a future that the integration density of individual electronic devices is increased by constantly decreasing minimum feature dimensions.

(5) Currently, the advanced packaging methods include: Wafer Level Chip Scale Packaging (WLCSP), Fan-Out Wafer Level Package (FOWLP), FlipChip, Package on Package (POP), and Fan-out Panel level Package (FOPLP). The fan-out package has unique advantages over the conventional wafer level package: (1) having flexible I/O spacing, independent of chip dimension; (2) only using active die, thereby improving the product yield; (3) having a flexible 3D packaging path, namely, making it possible to form an arbitrary array of patterns at the top; (4) having good electrical performance and thermal performance; (5) high frequency applications; and (6) being easy to realize high-density wiring in a redistribution layer (RDL). The fan-out packaging method is generally: providing a carrier substrate, and forming an adhesive layer on the surface of the substrate; performing photolithography and electroplating to obtain redistribution layers (RDL) on the adhesive layer; mounting a semiconductor chip on the redistribution layers by using a chip bonding process; using an injection molding process to plastically package a semiconductor chip in a plastic packaging material layer; removing the substrate and the adhesive layer; performing photolithography and electroplating on the redistribution layers to form an under bump metallurgy layer (UBM); and performing ball mount reflow on the UBM to form a solder bump.

(6) The panel level package in fan-out package accommodates several times as many components and parts as wafer level due to the larger substrate dimension compared with the wafer level package, which is more cost-effective in meeting the same competition in process capability.

(7) The fan-out packaging structure has the defects of insufficient input/output ports, large dimensions of packaging structure, etc. How to address the insufficient number of input/output ports in a panel level fan-out package and how to reduce product dimension become key points.

(8) In the prior art, CN111106013A discloses a preparation method of TMV (Through Molding Via) structure, and a large-board fan-out type heterogeneous integrated packaging structure. As shown in FIG. 1, the packaging structure uses a board level technology to make a chip plastic package plate **10**, uses a laser electroplating method to make a TMV structure **12**, provides an antenna **13** and a surface-mount element **11** and makes a connection circuit **15** to be connected, and makes a redistribution layer **16** and a solder ball **17** on the other side of the plastic package plate.

(9) However, the laser or drilling process technology of the prior art has low processing efficiency and high production cost; whether the TMV is made with laser or machining drilling related process, it has poor machining accuracy, and singular structure, shape, and dimension, and the TMV structure with large dimension and multi-shape cannot be well handled; in laser porefilling electroplating, the bonding force between the copper pillar and the dielectric layer is poor (the outer surface of the copper pillar cannot be treated), which affects the product reliability; in addition, the

copper pillar by laser porefilling electroplating is tapered from top to bottom, and the difference in the longitudinal dimension of the copper pillar is not conducive to heat dissipation and signal transmission and stability.

SUMMARY

(10) Embodiments of the present invention involve providing an embedded packaging structure and a manufacturing method thereof to solve the above technical problems. According to the present invention, devices are implanted after the circuit layout, wherein the devices can be selectively implanted according to the yield of the substrate, so as to reduce the loss of the devices; the devices, by a POP back-to-back stacked manner, increase the I/O number and reduce the area ratio when the devices are arranged horizontally; the pattern of the circuit and the copper pillar can be set arbitrarily according to practical requirements, and can be realized only by electroplating the pattern image; the upper and lower dimensions of the circuit and the copper pillar are uniform, which is more advantageous for the heat dissipation of the package and the signal transmission stability; the surface of the copper pillar is browned after electroplating the copper pillar, which increases the bonding force between the copper pillar and the dielectric layer.

(11) The first aspect of the present invention relates to a multilayer embedded packaging structure, comprising a first dielectric layer and a second dielectric layer on the first dielectric layer, wherein the first dielectric layer comprises a first wiring layer, and the second dielectric layer comprises a first copper pillar layer penetrating through the second dielectric layer in a height direction, a device placement port frame, and a second wiring layer on the first copper pillar layer; a second copper pillar layer is provided on the second wiring layer, and the first wiring layer and the second wiring layer are conductively connected via the first copper pillar layer; a first device is mounted to a bottom of the device placement port frame such that a terminal of the first device is in conductive connection with the first wiring layer, a second device is mounted to the second dielectric layer such that the terminal of the second device is in conductive connection with the second wiring layer, and a third device is mounted to an end of the second copper pillar layer such that the terminal of the third device is in conductive connection with the second copper pillar layer.

(12) In some embodiments, gaps between the first device and the second device, and the packaging structure are filled with a dielectric material to form a plastic package layer.

(13) In some embodiments, the first dielectric layer and the second dielectric layer comprise an organic dielectric material, an inorganic dielectric material, or a combination thereof. Preferably, the first dielectric layer and the second dielectric layer comprise polyimide, epoxy resin, bismaleimide triazine resin, ceramic filler, glass fiber, or a combination thereof.

(14) In some embodiments, the third device is covered with a third dielectric layer.

(15) In some embodiments, the first device and the second device comprise a device having a double-sided terminal such that the terminal of the second device is in conductive connection with the terminal of the first device and the terminal of the second device is in conductive connection with the terminal of the third device.

(16) Preferably, the first device, the second device, and the third device comprise at least one device respectively.

(17) In some embodiments, a solder mask and a solder resist window are provided on the bottom surface of the first dielectric layer.

(18) The second aspect of the present invention provides a manufacturing method for a multilayer embedded packaging structure, comprising steps of: (a) forming a first wiring layer on a temporary carrier plate, laminating a first dielectric layer on the first wiring layer, and thinning the first dielectric layer to expose the above-mentioned first wiring layer; (b) forming a first copper pillar layer on the first dielectric layer, the first copper pillar layer comprising a sacrificial copper pillar, laminating a second dielectric layer on the first copper pillar layer, and thinning the second dielectric layer to expose the first copper pillar layer; (c) forming a second wiring layer on the second dielectric layer, such that the first wiring layer and the second wiring layer are in conductive

connection through the first copper pillar layer; (d) forming a second copper pillar layer on the second wiring layer; (e) etching the sacrificial copper pillar to form a device placement port frame exposing the first wiring layer; and (f) removing the temporary carrier plate.

(19) In some embodiments, the manufacturing method further includes: (g) mounting a first device on the bottom of the device placement port frame such that a terminal of the first device is in conductive connection with the first wiring layer; (h) mounting a second device on the second wiring layer such that the terminal of the second device is in conductive connection with the second wiring layer; and (i) mounting a third device on an end of the second copper pillar layer such that the terminal of the third device is in conductive connection with the second copper pillar layer.

(20) Preferably, the manufacturing method further comprises: (h') after step h and before step i, filling a dielectric material to form a plastic package layer covering the first device and the second device.

(21) Preferably, the manufacturing method further comprises: (i') after step i, laminating a dielectric material to form a plastic package layer covering the third device.

(22) In some embodiments, step (a) further comprises: (a1) applying a first photoresist layer on the temporary carrier plate, and performing exposure and development to form a first feature pattern; (a2) performing electroplating in the first feature pattern to form a first wiring layer and removing the first photoresist layer; and (a3) laminating a first dielectric layer on the first wiring layer, and thinning the first dielectric layer to expose the first wiring layer.

(23) Preferably, the temporary carrier plate comprises a double-sided copper-clad plate, wherein the double-sided copper-clad plate comprises a prepreg, a first copper layer on a surface of the prepreg, and a second copper layer on the first copper layer, the first copper layer and the second copper layer being attached together by physical press-fitting. Preferably, step (e) comprises removing the double-sided copper-clad plate by physically separating the first copper layer and the second copper layer and etching the second copper layer.

(24) In some embodiments, step (f) further comprises: (b1) forming a metal seed layer on the first dielectric layer; (b2) applying a second photoresist layer on the metal seed layer, and exposing and developing to form a second feature pattern; (b3) performing electroplating in the second feature pattern to form an etch resist layer; (b4) applying a third photoresist layer, and exposing and developing to form a third feature pattern; (b5) performing electroplating in the third feature pattern to form a first copper pillar layer, and a sacrificial copper pillar on the etch resist layer, and removing the second photoresist layer and the third photoresist layer; and (b6) laminating a second dielectric layer over the first feature layer and the sacrificial copper pillar, and thinning the second dielectric layer to expose the first feature layer and the sacrificial copper pillar.

(25) In some embodiments, the etch resist layer comprises nickel, titanium, or a combination thereof. In some embodiments, step (c) further comprises: (c1) forming a metal seed layer on the second dielectric layer; (c2) applying a fourth photoresist layer on the metal seed layer, and exposing and developing to form a fourth feature pattern; and (c3) electroplating copper in the fourth feature pattern to form a second wiring layer, and removing the fourth photoresist layer.

(26) In some embodiments, step (d) comprises: (d1) applying a fifth photoresist layer on the second wiring layer, and performing exposure and development to form a fifth feature pattern; (d2) electroplating copper in the fifth feature pattern to form a second copper pillar layer; and (d3) removing the fifth photoresist layer and etching the exposed metal seed layer. Preferably, the metal seed layer comprises titanium, copper, a titanium-tungsten alloy, or a combination thereof.

(27) In some embodiments, step (d) further comprises: (d1) applying a sixth photoresist layer on the second wiring layer, and exposing and developing to form a sixth feature pattern; and (d2) etching the sacrificial copper pillar and the etch resist layer in the sixth feature pattern to form a device placement port frame.

(28) In some embodiments, the manufacturing method further comprises after removing the

temporary carrier plate in step f, applying a solder resist layer on a bottom surface of the first dielectric layer and surface treating exposed metal to form a solder resist window.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a better understanding of the present invention and to show the implementation mode thereof, reference will now be made, purely by way of example, to the accompanying drawings.
- (2) With specific reference to the accompanying drawings, it must be stressed that the particular graphical representations shown are by way of example and for purposes of illustrative discussion of preferred embodiments of the present invention only, and are presented in the cause of providing what is believed to be the most useful and readily understood graphical representations of the description of the principles and conceptual aspects of the present invention. In this regard, no attempt is made to show structural details of the present invention in more details than is necessary for a fundamental understanding of the present invention; the description with reference to the accompanying drawings makes it apparent to those skilled in the art how several forms of the present invention may be embodied in practice.
- (3) In the drawings:
- (4) FIG. 1 is a schematic cross-sectional view of a preparation method for a TMV structure and a large plate fan-out heterogeneous integrated packaging structure in the prior art;
- (5) FIG. 2 is a schematic cross-sectional view of an embedded packaging structure according to one embodiment of the present invention;
- (6) FIG. 3 is a schematic cross-sectional view of an embedded packaging structure according to another embodiment of the present invention;
- (7) FIG. 4 is a schematic cross-sectional view of an embedded packaging structure according to another embodiment of the present invention;
- (8) FIG. 5 is a schematic cross-sectional view of an embedded packaging structure according to another embodiment of the present invention;
- (9) FIGS. 6A-6P show schematic cross-sectional views of an intermediate structure at various steps of a method for manufacturing a packaging structure according to one embodiment of the present invention.

DETAILED DESCRIPTION

- (10) Referring to FIG. 2, a schematic cross-sectional view of an embedded packaging structure **100** is shown. The packaging structure **100** includes a first dielectric layer **101** and a second dielectric layer **102** on the first surface of the first dielectric layer **101**. The first dielectric layer **101** and the second dielectric layer **102** may comprise the same material or different materials; and may comprise organic dielectric materials, inorganic dielectric materials or combinations thereof, preferably, polyimides, epoxy resins, bismaleimide triazine resins (BT), ceramic filler, glass fibers, or combinations thereof.
- (11) The first dielectric layer **101** comprises a first wiring layer **1013**, the first wiring layer **1013** is exposed on the first surface of the first dielectric layer **101**, the second dielectric layer **102** comprises a first through-hole pillar **1024** penetrating the second dielectric layer **102** in a height direction and a device placement port frame **1026**, the second dielectric layer **102** is provided thereon with a second wiring layer **1032**, the second wiring layer **1032** is provided thereon with a second through-hole pillar **1033**, and the first wiring layer **1013** and the second wiring layer **1032** are conductively connected via the first through-hole pillar **1024**. The second dielectric layer **102** includes a first copper pillar layer **1024**, which serves as an IO channel for the copper through-hole pillar and may have the same or different cross-sectional dimensions. The second copper pillar layer **1033** also serves as an IO channel. Arranging the wiring layer before implanting a device

allows selective implantation according to the yield of the substrate, thereby reducing device losses.

(12) The bottom of the device placement port frame **1026** is mounted with a first device **1051**, the second wiring layer **1032** is mounted thereon with a second device **1052**, the end of the second copper pillar layer **1033** is mounted with a third device **1053**, and the gap between the first device **1051** and the second device **1052** and the structure **100** can be filled with a dielectric material to form a plastic package layer. With the back-to-back stacked arrangement adopted by the first device **1051**, the second device **1052**, and the third device **1053**, the number of I/O per unit area can be significantly increased, thereby reducing the area ratio when the devices are arranged horizontally.

(13) As shown in FIG. 2, the packaging structure **100** further includes a solder resist layer **106** formed on the second surface of the first dielectric layer **101**, a solder resist window **1061** being provided in the solder resist layer **106**.

(14) Referring to FIG. 3, a schematic cross-sectional view of an embedded packaging structure **200** is shown. The difference between the packaging structure **200** and the packaging structure **100** is that the first device **1051** and the second device **1052** each has a double-sided terminal, wherein the terminal of the first device **1051** is in conductive connection with the terminal of the second device **1052**, and the terminal of the second device **1052** is in conductive connection with the terminal of the third device **1053**.

(15) Referring to FIG. 4, a schematic cross-sectional view of an embedded packaging structure **300** is shown. The packaging structure **300** differs from the packaging structure **100** in that a third dielectric layer **103** is provided on the second dielectric layer **102**, and the third dielectric layer **103** covers the second wiring layer **1032** and a second through-hole pillar **1033**; a fourth dielectric layer **104** is provided on the third dielectric layer **103**, and the fourth dielectric layer **104** covers the third device **1053**. The third dielectric layer **103** and the first dielectric layer **101** may comprise the same material or different materials; the fourth dielectric layer **104** and the first dielectric layer **101** may comprise the same material or different materials. Preferably, the fourth dielectric layer **104** is selected from an organic dielectric material or a photosensitive dielectric material, for example, the organic dielectric material may be ABF or PP, and the photosensitive dielectric material may be PID.

(16) Referring to FIG. 5, a schematic cross-sectional view of an embedded packaging structure **400** is shown. The difference between the packaging structure **400** and the packaging structure **300** is that the first device **1051** and the second device **1052** each has a double-sided terminal, wherein the terminal of the first device **1051** is in conductive connection with the terminal of the second device **1052**, and the terminal of the second device **1052** is in conductive connection with the terminal of the third device **1053**.

(17) Referring to FIGS. 6A-6P, schematic cross-sectional views of an intermediate structure at various steps of a method for manufacturing a multilayer embedded packaging structure according to one embodiment of the present invention are shown.

(18) The manufacturing method comprises the steps of: preparing a temporary carrier plate, such as a double-sided copper-clad plate DTF **10**, applying a first photoresist layer **1012** on at least one side of the DTF **10**, and exposing and developing to form a first feature pattern, step (a), as shown in FIG. 6A. The DTF **10** comprises a PP (prepreg) layer **1011a**, and the PP layer **1011a** is successively a first copper layer **1011b** located on the surface of the PP layer **1011a**, a second copper layer **1011c** located on the surface of the first copper layer **1011b**, and a protective layer **1011d** located on the surface of the second copper layer **1011c** in an outward way; the first copper layer **1011b** and the second copper layer **1011c** are formed by physically press-fitting copper foils, and the first copper layer **1011b** and the second copper layer **1011c** can be physically separated, facilitating the removal of the DTF10 in the subsequent process; the protective layer **1011d** may comprise copper, titanium, nickel, tungsten or a combination thereof; the thickness of the first

copper layer **1011b**, the second copper layer **1011c**, and the protective layer **1011d** can be adjusted according to practical requirements, and preferably, the thickness of the first copper layer **1011b** is 18 μm , the thickness of the second copper layer **1011c** is 3 μm , and the thickness of the protective layer **1011d** is 3-10 μm . Generally, the first photoresist layer **1012** may be applied simultaneously to two sides of the DTF10 and exposed and developed to form a first feature pattern. In this embodiment, only a single unit on one side of the DTF10 is subsequently demonstrated, but it is not limited to performing subsequent operations on only one side of the DTF10.

(19) Next, in the pattern, copper is electroplated to form the first wiring layer **1013**, the first photoresist layer **1012** is removed, the dielectric material is laminated, and the dielectric material is thinned to expose the surface of the first wiring layer **1013** to form the first dielectric layer **101**, step (b), as shown in FIG. 6B. Generally, in order to ensure a filling effect, the amount of the dielectric material is calculated based on the filling amount, and preferably, the dielectric material is 5-20 μm higher than the surface of the first wiring layer **1013**; the dielectric material may comprise an organic dielectric material, an inorganic dielectric material, or a combination thereof, and preferably, the dielectric material comprises polyimide, epoxy resin, bismaleimide triazine resin, ceramic filler, glass fiber, or a combination thereof. Being divided according to functional requirements, the dielectric material may be selected from a photosensitive material and a non-photosensitive material, and preferably, the dielectric material adopts a photosensitive dielectric material. The dielectric material can be thinned integrally by means of grinding plate or plasma etching, and can also be thinned locally by means of laser or drilling; when the dielectric material adopts a photosensitive material, the dielectric material can also be locally thinned by means of exposure and development.

(20) Then, a metal seed layer **1020** is formed on the first surface of the first dielectric layer **101**, a second photoresist layer **1021** is applied on the metal seed layer **1020**, via exposure and development, a second feature pattern is formed, and electroplating is performed in the pattern to form an etch resist layer **1022**, step (c), as shown in FIG. 6C. Generally, the etch resist layer **1022** can comprise nickel, titanium, or a combination thereof, for example, an 8-15 μm nickel layer. Generally, the metal seed layer **1020** can be formed on the first dielectric layer **101** by means of electroless plating or sputtering, wherein the metal seed layer **1020** can comprise titanium, copper, a titanium-tungsten alloy, or a combination thereof, and the thickness of the metal seed layer **1020** ranges from 0.8 μm to 5 μm ; preferably, the metal seed layer **1020** is made by sputtering with 0.1 μm titanium and 1 μm copper.

(21) Next, a third photoresist layer **1023** is applied, and exposure and development are performed to form a third feature pattern in which copper is electroplated to form a first copper pillar layer **1024** and a sacrificial copper pillar **1025**, step (d), as shown in FIG. 6D. Generally, after the first copper pillar layer **1024** is formed, a browning treatment can be performed on the surface of the first copper pillar layer **1024**, so as to increase the bonding force between the first copper pillar layer **1024** and the dielectric layer coated on the outside thereof; the shape of the first copper pillar layer **1024** may be set according to actual requirements, for example, it may be square, circular, etc. without limitation. The thickness of the sacrificial copper pillar **1025** may be adjusted according to the thickness of the device. The first copper pillar layer **1024** has uniform upper and lower dimensions, which is more advantageous for heat dissipation and signal transmission stability of the embedded packaging structure. The sacrificial copper pillar **1025** is located on the surface of the etch resist layer **1022** such that the etch resist layer **1022** can protect the first wiring layer **1013** from etching when the sacrificial copper pillar **1025** is subsequently etched.

(22) Then, the second photoresist layer **1021** and the third photoresist layer **1023** are removed, the dielectric material is laminated, and the dielectric material is thinned to expose the first copper pillar layer **1024** and the sacrificial copper pillar **1025** to form the second dielectric layer **102**, step (e), as shown in FIG. 6E.

(23) Next, a second wiring layer **1032** is formed on the second dielectric layer **102**, step (f), as

shown in FIG. 6F. Generally, the following steps are included: forming a metal seed layer **1031** on the second dielectric layer **102**; applying a fourth photoresist layer **1034** on the metal seed layer **1031**, and performing exposure and development to form a fourth feature pattern; and electroplating copper in a pattern to form a second wiring layer **1032**.

(24) Generally, the metal seed layer **1031** can be formed on the second dielectric layer **102** by means of electroless plating or sputtering, wherein the metal seed layer **1031** can comprise titanium, copper, a titanium-tungsten alloy, or a combination thereof, and the thickness of the metal seed layer **1031** ranges from 0.8 μm to 5 μm ; preferably, the metal seed layer **1031** is made by sputtering with 0.1 μm titanium and 1 μm copper.

(25) Then, a second copper pillar layer **1033** is formed on the second wiring layer **1032**, step (g), as shown in FIG. 6G. Generally, the following steps are included: applying a fifth photoresist layer, and exposing and developing to form a fifth feature pattern; electroplating copper in a pattern to form a second copper pillar layer **1033**; and removing the fourth photoresist layer **1034** and the fifth photoresist layer, and etching the exposed metal seed layer **1031**.

(26) Generally, the shape of the second copper pillar layer **1033** can be set according to practical requirements, for example, it can be square, circular, etc. without specific limitation; the second copper pillar layer **1033** has uniform upper and lower dimensions, which is more advantageous for heat dissipation and signal transmission stability of the embedded packaging structure. After the second copper pillar layer **1033** is formed by electroplating, a browning treatment may also be performed on the surface of the second copper pillar layer **1033** so as to increase the bonding force between the second copper pillar layer **1033** and the dielectric layer coated on the outside thereof.

(27) Next, a sixth photoresist layer **1036** is applied, exposure and development are performed to form a sixth feature pattern, the sacrificial copper pillar **1025** is etched and the protective layer **1022** is etched to form a device placement port frame **1026**, step (h), as shown in FIG. 6H.

(28) Generally, the sixth photoresist layer **1036** can be applied on the second copper pillar layer **1033** and the second wiring layer **1032** to protect the second copper pillar layer **1033** and the second wiring layer **1032** from being etched when the sacrificial copper pillar **1025** is subsequently etched; the sixth photoresist layer **1036** is exposed and developed to expose the sacrificial copper pillar **1025**, the sacrificial copper pillar **1025** is etched and the protective layer **1022** is etched.

(29) Then, the whole plate is applied with a seventh photoresist layer and is exposed and cured, the first copper layer **1011b** and the second copper layer **1011c** are separated, the second copper layer **1011c** and the protective layer **1011d** are etched, and the sixth photoresist layer **1036** and the seventh photoresist layer are removed to form a substrate, step (i), as shown in FIG. 6I. The seventh photoresist layer is applied to protect the substrate during the removal of the DTF10. The present invention may merely provide a substrate provided with a wiring layer and a through-hole pillar, and then a packaging manufacturer may further mount a device, and then package the same, thereby eliminating process equipment input such as pattern electroplating for the subsequent packaging manufacturer.

(30) Next, proceeding step (i), a first device **1051** is mounted on the bottom of the device placement port frame **1026**, a second device **1052** is mounted on the surface of the second wiring layer **1032**, and a third device **1053** is mounted on the end of the second copper pillar layer **1033**, step (j), as shown in FIG. 6J. Generally, each device has two terminal faces corresponding to each other, or has a single-sided terminal face and a non-terminal face corresponding to the single-sided terminal face, and the device can realize the electrical signal connection between the terminal and the substrate by solder. A device with a single-sided terminal face is taken as an example to perform subsequent process description; when mounting a device, in one case, solder can be applied by printing or solder dispensing on the bottom of the device placement port frame **1026**, the surface of the second wiring layer **1032**, and the end of the second copper pillar layer **1033**, and then the device is mounted, and then reflow soldering is performed; in one case, solder may be affixed to the terminal surfaces of the first device **1051**, the second device **1052**, and the third device **1053**,

and they are mounted to the bottom of the device placement port frame **1026**, the surface of the second wiring layer **1032**, and the end of the second copper pillar layer **1033**, respectively, and then reflow soldering is performed. The solder may include solder paste and silver paste. The devices are implanted after the circuit layout, wherein the devices can be selectively implanted according to the yield of the substrate, so as to reduce the loss of the devices; the devices are implanted in a back-to-back stacked manner, increasing the I/O number and reducing the area ratio when the devices are arranged horizontally.

(31) The dielectric material is then filled to fill the gap between the first device **1051** and the device placement port frame **1026**, step (k), as shown in FIG. **6K**. Generally, the dielectric material may be filled downward from the direction of mounting the device. Preferably, the dielectric material is selected from an organic dielectric material or a photosensitive dielectric material, for example, the organic dielectric material may be ABF or PP, and the photosensitive dielectric material may be PID.

(32) Next, a solder mask **106** is applied to the second surface of the first dielectric layer **101**, and the exposed metal is subjected to a metal surface treatment to form the solder resist window **1061**, step (1), as shown in FIG. **6L**. Generally, the metal surface treatment may be performed by chemical nickel-palladium gold, or OPS, or the like. If the first device **1051** and the second device **1052** have double-sided terminals, the embedded packaging structure is as shown in FIG. **3**. A part of the terminal of the first device **1051** and a part of the terminal of the second device **1052** are soldered, and a part of the terminal of the second device **1052** and a part of the terminal of the third device **1053** are soldered.

(33) Then, proceeding to step (i), the first device **1051** is mounted on the bottom of the device placement port frame **1026**, and the second device **1052** is mounted on the surface of the second wiring layer **1032**, step (m), as shown in FIG. **6M**.

(34) Next, the dielectric material is laminated to cover the second wiring layer **1032**, the second copper pillar layer **1033**, and the second device **1052**, and the dielectric material is thinned to expose ends of the second copper pillar layer **1033** to form a third dielectric layer **103**, step (n), as shown in FIG. **6N**. Generally, the dielectric material can be laminated downward from the direction of mounting the device. In order to ensure a filling effect, the dielectric material is generally at least 5 μm higher than the end of the second copper pillar layer **1033**; preferably, the dielectric material is selected from an organic dielectric material or a photosensitive dielectric material, for example, the organic dielectric material may be ABF or PP, and the photosensitive dielectric material may be PD.

(35) Then, a third device **1053** is mounted to the end of the second copper pillar layer **1033**, and the third device **1053** is plastic packaged with the dielectric material to form a fourth dielectric layer **104**, step (o), as shown in FIG. **6O**.

(36) Finally, the solder mask **106** is applied on the second surface of the first dielectric layer **101** and the exposed metal is subjected to metal surface treatment to form the solder resist window **1061**, step (p), as shown in FIG. **6P**. If the first device **1051** and the second device **1052** have double-sided terminals, the embedded packaging structure is as shown in FIG. **5**. A part of the terminal of the first device **1051** and a part of the terminal of the second device **1052** are soldered, and a part of the terminal of the second device **1052** and a part of the terminal of the third device **1053** are soldered.

(37) It will be appreciated by technicians skilled in the art that the present invention is not limited by what has been particularly shown and described in the context. Rather, the scope of the present invention is defined by the appended claims, including both combinations and sub-combinations of the various technical features described hereinabove, as well as variations and modifications thereof, which would occur to technicians skilled in the art upon reading the foregoing description.

(38) In the claims, the term “comprises” and variations thereof such as “comprising”, “which

comprises” and the like mean that the recited assembly is included, but generally not excluding other assemblies.

Claims

1. A method for manufacturing a multilayer embedded packaging structure, the method comprising: (a) forming a first wiring layer on a temporary carrier plate, laminating a first dielectric layer on the first wiring layer, and thinning the first dielectric layer to expose the above-mentioned first wiring layer; (b) forming a first copper pillar layer on the first dielectric layer, the first copper pillar layer comprising a sacrificial copper pillar, laminating a second dielectric layer on the first copper pillar layer, and thinning the second dielectric layer to expose the first copper pillar layer, the forming of the first copper pillar layer comprising: (b1) forming a metal seed layer on the first dielectric layer; (b2) applying a second photoresist layer on the metal seed layer of the first dielectric layer, and exposing and developing to form a second feature pattern; (b3) performing electroplating in the second feature pattern to form an etch resist layer; (b4) applying a third photoresist layer, and exposing and developing to form a third feature pattern; (b5) performing electroplating in the third feature pattern to form a first copper pillar layer, and a sacrificial copper pillar on the etch resist layer, and removing the second photoresist layer and the third photoresist layer; and (b6) laminating the second dielectric layer over the first feature layer and the sacrificial copper pillar, and thinning the second dielectric layer to expose the first feature layer and the sacrificial copper pillar; (c) forming a second wiring layer on the second dielectric layer such that the first wiring layer and the second wiring layer are in conductive connection through the first copper pillar layer; (d) forming a second copper pillar layer on the second wiring layer; (e) etching the sacrificial copper pillar to form a device placement port frame exposing the first wiring layer; and (f) removing the temporary carrier plate.
2. The manufacturing method according to claim 1, further comprising: (g) mounting a first device on a bottom of the device placement port frame such that a terminal of the first device is in conductive connection with the first wiring layer; (h) mounting a second device on the second wiring layer such that the terminal of the second device is in conductive connection with the second wiring layer; and (i) mounting a third device on an end of the second copper pillar layer such that the terminal of the third device is in conductive connection with the second copper pillar layer.
3. The manufacturing method according to claim 2, further comprising: (h') after step h and before step i, filling a dielectric material to cover the first device and the second device.
4. The manufacturing method according to claim 2, further comprising: (i') after step i, laminating a dielectric material to form a plastic package layer covering the third device.
5. The manufacturing method according to claim 1, wherein step (a) comprises: (a1) applying a first photoresist layer on the temporary carrier plate, and performing exposure and development to form a first feature pattern; (a2) performing electroplating in the first feature pattern to form a first wiring layer and removing the first photoresist layer; and (a3) laminating a first dielectric layer on the first wiring layer, and thinning the first dielectric layer to expose the first wiring layer.
6. The manufacturing method according to claim 1, the temporary carrier plate comprising a double-sided copper-clad plate, wherein the double-sided copper-clad plate comprises a prepreg, a first copper layer on a surface of the prepreg, and a second copper layer on the first copper layer, the first copper layer and the second copper layer being attached together by physical press-fitting.
7. The manufacturing method according to claim 6, wherein step (e) comprises removing the double-sided copper-clad plate by physically separating the first copper layer and the second copper layer and etching the second copper layer.
8. The manufacturing method according to claim 1, wherein the etch resist layer comprises nickel, titanium, or a combination thereof.
9. The manufacturing method according to claim 1, wherein step (c) comprises: (c1) forming a

metal seed layer on the second dielectric layer; (c2) applying a fourth photoresist layer on the metal seed layer of the second dielectric layer, and exposing and developing to form a fourth feature pattern; and (c3) electroplating copper in the fourth feature pattern to form the second wiring layer, and removing the fourth photoresist layer.

10. The manufacturing method according to claim 1, wherein step (d) comprises: (d1) applying a fifth photoresist layer on the second wiring layer, and performing exposure and development to form a fifth feature pattern; (d2) electroplating copper in the fifth feature pattern to form a second copper pillar layer; and (d3) removing the fifth photoresist layer.

11. The manufacturing method according to claim 1, wherein the metal seed layer comprises titanium, copper, a titanium-tungsten alloy, or a combination thereof.

12. The manufacturing method according to claim 1, wherein step (d) comprises: (d1) applying a sixth photoresist layer on the second wiring layer, and exposing and developing to form a sixth feature pattern; and (d2) etching the sacrificial copper pillar and the etch resist layer in the sixth feature pattern to form a device placement port frame.

13. The manufacturing method according to claim 1, further comprising after removing the temporary carrier plate in step f, applying a solder resist layer on a bottom surface of the first dielectric layer and surface treating exposed metal to form a solder resist window.
